Quantum Computing

Fourier Methods, Phase Estimation, and Shor's Algorithm

Lecture 7 Notes

Ralph Razzouk Marc Fares Joseph Jabbour

Lebanese Science Organization

Prepared for the LebNet and LebSci Quantum Computing Course 2026

About these notes

These notes accompany a nine-lecture course on quantum computing designed and produced by the Lebanese Science Organization. Each chapter corresponds to one lecture. The lectures are concept-focused: they explain what a quantum state is, how entanglement, interference, and measurement combine to produce computation, and what is required to make that computation reliable. Linear algebra, complex numbers, and elementary probability are assumed.

Each chapter names its lecturer and date, opens with a short preface, develops its material with the derivations in place, and closes with a summary. Worked examples appear in numbered boxes, as do definitions, theorems, remarks, and common misconceptions. Works cited in the text are collected in a single reference list at the end of the document.

Conventions used throughout: qubits in a register are labelled from left to right, so that in $|ab\rangle$ the first qubit is a; the first qubit is the top wire in every circuit diagram; the computational basis of two qubits is ordered $|00\rangle, |01\rangle, |10\rangle, |11\rangle$; circuits are read from left to right, so a gate U_1 followed by a gate U_2 acts as the matrix product U_2U_1.

This chapter is exported individually from the full lecture notes (7 of 9). These notes are made available for open, public study. Reuse conditions will be stated separately by the Lebanese Science Organization.

Contents

1 Fourier Methods, Phase Estimation, and Shor's Algorithm **3**
1.1 The quantum Fourier transform . . . 3
1.1.1 Orthonormality . . . 4
1.1.2 The two-qubit transform . . . 4
1.1.3 The product form . . . 5
1.1.4 Circuit structure . . . 5
1.1.5 The three-qubit circuit . . . 6
1.2 Phase estimation . . . 6
1.2.1 Exact eigenphases . . . 7
1.2.2 Inexact eigenphases . . . 8
1.3 Order finding and factoring . . . 9
1.3.1 Eigenstates of modular multiplication . . . 10
1.3.2 The classical reduction . . . 10
1.3.3 The instance $N = 15$ . . . 11
1.3.4 Shor's algorithm . . . 11
1.4 Summary . . . 12

References **13**

Chapter 1

Fourier Methods, Phase Estimation, and Shor's Algorithm

Lecturer: Joseph Jabbour

Date: November 7, 2026

A unitary leaves each of its eigenstates unchanged except for a phase, and a phase is invisible to a direct measurement. This lecture develops the machinery that converts such a phase into bits. The quantum Fourier transform extends the Hadamard gate from signs to arbitrary phases; phase estimation applies controlled powers of a unitary and the inverse Fourier transform to write an eigenphase into a register that a measurement can read. Shor's algorithm then reduces factoring to the period of modular exponentiation, which is an eigenphase problem of exactly this kind, and we account for what the reduction costs.

1.1 The quantum Fourier transform

In **??** a single Hadamard gate turned the relative sign between two amplitudes into a bit that a computational-basis measurement could read. In **??** the gate $H^{\otimes n}$ did the same for a pattern of signs ± 1 spread over 2^n amplitudes, and the Deutsch-Jozsa algorithm rested on that conversion. This lecture introduces the operation that performs the conversion for arbitrary phases rather than signs. It is the quantum Fourier transform, and the Hadamard gate is its one-qubit case.

Throughout, a register of n qubits is described as in **??**. The computational basis states are labelled by integers $x \in \{0, 1, \ldots, N-1\}$ with $N = 2^n$, and the binary expansion $x = x_1 x_2 \cdots x_n$ has its most significant bit x_1 on the first qubit, which is the top wire of every circuit.

Definition 1.1: Quantum Fourier transform

The quantum Fourier transform on n qubits, with $N = 2^n$, is the linear map defined on the computational basis by

$$\mathrm{QFT}\,|x\rangle = \frac{1}{\sqrt{N}} \sum_{y=0}^{N-1} \mathrm{e}^{2\pi \mathrm{i} x y/N}\,|y\rangle\,, \qquad x = 0, 1, \ldots, N-1. \tag{1.1}$$

Its matrix in the computational basis has entries $\mathrm{QFT}_{yx} = \omega^{xy}/\sqrt{N}$ with $\omega = \mathrm{e}^{2\pi \mathrm{i}/N}$.

For $n = 1$ we have $N = 2$ and $\omega = -1$, and (1.1) reads $\mathrm{QFT}\,|0\rangle = |+\rangle$ and $\mathrm{QFT}\,|1\rangle = |-\rangle$. The quantum Fourier transform on one qubit is the Hadamard gate. For $n = 2$ we have $\omega = \mathrm{i}$, and the 4×4 matrix is written out in section 1.1.2.

The map QFT is unitary: its columns are orthonormal, by a geometric sum carried out in section 1.1.1. It is therefore a legitimate gate, and its inverse $\mathrm{QFT}^{-1} = \mathrm{QFT}^{\dagger}$ is obtained by replacing $\mathrm{e}^{2\pi \mathrm{i} xy/N}$ with $\mathrm{e}^{-2\pi \mathrm{i} xy/N}$ in (1.1). Being unitary, it is also a change of orthonormal basis, and it is useful to name the basis it produces.

Definition 1.2: Fourier basis

The Fourier basis of the n-qubit register is the orthonormal set $\{|f_x\rangle : x = 0, \ldots, N-1\}$ with $|f_x\rangle = \mathrm{QFT}\,|x\rangle$. Each $|f_x\rangle$ has every computational-basis amplitude of modulus $1/\sqrt{N}$; the states differ from one another only in the pattern of relative phases $\mathrm{e}^{2\pi \mathrm{i} xy/N}$ across the register.

Two Fourier basis states are indistinguishable by a computational-basis measurement, exactly as $|+\rangle$ and $|-\rangle$ were in **??**. Applying QFT^{-1} and then measuring distinguishes them perfectly, since $\mathrm{QFT}^{-1}\,|f_x\rangle = |x\rangle$. This is the whole use we make of the transform: a computation that succeeds in placing the register in a Fourier basis state, or close to one, has encoded an integer x in relative phases, and QFT^{-1} followed by measurement reads x out. The next section shows how phase kickback places a register in such a state.

1.1.1 Orthonormality

The inner product of two columns of QFT is

$$\langle f_{x'}|f_x\rangle = \frac{1}{N}\sum_{y=0}^{N-1} \mathrm{e}^{-2\pi \mathrm{i} x'y/N}\mathrm{e}^{2\pi \mathrm{i} xy/N} = \frac{1}{N}\sum_{y=0}^{N-1} \omega^{(x-x')y}, \qquad \omega = \mathrm{e}^{2\pi \mathrm{i}/N}. \tag{1.2}$$

For $x = x'$ every term is 1 and the sum is N, so $\langle f_x|f_x\rangle = 1$. For $x \neq x'$ set $q = \omega^{x-x'}$; then $q \neq 1$ because $0 < |x-x'| < N$, while $q^N = 1$, and the geometric sum gives $\sum_{y=0}^{N-1} q^y = (q^N - 1)/(q-1) = 0$. The columns are orthonormal, so $\mathrm{QFT}^{\dagger}\mathrm{QFT} = \mathbb{I}$ and QFT is unitary. The same calculation shows that $\mathrm{QFT}^{-1} = \mathrm{QFT}^{\dagger}$ has entries $\omega^{-xy}/\sqrt{N}$.

1.1.2 The two-qubit transform

For $n = 2$, $N = 4$ and $\omega = \mathrm{e}^{2\pi \mathrm{i}/4} = \mathrm{i}$. In the basis order $|00\rangle, |01\rangle, |10\rangle, |11\rangle$, that is $x = 0, 1, 2, 3$, the entry in row y and column x is $\mathrm{i}^{xy}/2$:

$$\mathrm{QFT} = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & \mathrm{i} & -1 & -\mathrm{i} \\ 1 & -1 & 1 & -1 \\ 1 & -\mathrm{i} & -1 & \mathrm{i} \end{pmatrix}. \tag{1.3}$$

The first column is $|+\rangle\,|+\rangle$, and the third column is $|+\rangle\,|-\rangle$: both are product states, as the product form predicts. One checks directly that the columns are orthonormal; for instance the inner product of columns $x = 3$ and $x = 1$ is $\frac{1}{4}\sum_y \mathrm{i}^{-3y}\mathrm{i}^{y} = \frac{1}{4}\sum_y (-1)^y = 0$.

1.1.3 The product form

The definition (1.1) is a sum of $N = 2^n$ terms, and a naive implementation would be hopeless. The reason the transform is efficient is that the right-hand side factorizes. Writing $0.x_k x_{k+1} \cdots x_n$ for the binary fraction $x_k/2 + x_{k+1}/4 + \cdots + x_n/2^{n-k+1}$, one has the product form

$$\mathrm{QFT}\,|x_1 x_2 \cdots x_n\rangle = \bigotimes_{k=1}^{n} \frac{|0\rangle + \mathrm{e}^{2\pi \mathrm{i}\, 0.x_k x_{k+1} \cdots x_n}\,|1\rangle}{\sqrt{2}}, \tag{1.4}$$

which is derived below, first for $n = 2$ and then for general n. The factors in (1.4) are listed in reversed wire order: the factor of index k is the state of output qubit $n + 1 - k$, so the top wire carries the phase $0.x_n$ and the bottom wire carries $0.x_1 x_2 \cdots x_n$. We adopt this convention whenever the product form is used; the derivation checks it bit by bit.

Take $n = 2$ and write $y = 2y_1 + y_2$ with $y_1, y_2 \in \{0, 1\}$. Then $xy/4 = xy_1/2 + xy_2/4$. Since $x = 2x_1 + x_2$, the first term is $x_1 y_1 + x_2 y_1/2$, and the integer $x_1 y_1$ contributes nothing to $\mathrm{e}^{2\pi \mathrm{i} xy/4}$; so $\mathrm{e}^{2\pi \mathrm{i} x y_1/2} = \mathrm{e}^{2\pi \mathrm{i}\,(0.x_2)\,y_1}$. The second term is $xy_2/4 = (0.x_1 x_2)\,y_2$ exactly. Hence

$$\mathrm{QFT}\,|x_1 x_2\rangle = \frac{1}{2} \sum_{y_1, y_2} \mathrm{e}^{2\pi \mathrm{i}\,(0.x_2)\,y_1} \mathrm{e}^{2\pi \mathrm{i}\,(0.x_1 x_2)\,y_2}\,|y_1\rangle\,|y_2\rangle = \frac{|0\rangle + \mathrm{e}^{2\pi \mathrm{i}\, 0.x_2}\,|1\rangle}{\sqrt{2}} \otimes \frac{|0\rangle + \mathrm{e}^{2\pi \mathrm{i}\, 0.x_1 x_2}\,|1\rangle}{\sqrt{2}}. \tag{1.5}$$

The top wire carries the phase $0.x_2$ and the bottom wire carries $0.x_1 x_2$, in agreement with the convention stated after (1.4). For $x = 1$, that is $x_1 = 0$ and $x_2 = 1$, the factors are $|-\rangle$ and $(|0\rangle + \mathrm{i}\,|1\rangle)/\sqrt{2}$, whose product $\frac{1}{2}(|00\rangle + \mathrm{i}\,|01\rangle - |10\rangle - \mathrm{i}\,|11\rangle)$ is the second column of (1.3).

For general n the same steps apply. Writing $y = \sum_{l=1}^{n} y_l 2^{n-l}$, the exponent $xy/2^n = \sum_l y_l\, x/2^l$, and $x/2^l$ differs from the binary fraction $0.x_{n-l+1} \cdots x_n$ by an integer. The sum over y then factorizes into n independent sums over the bits y_l, giving on wire l the factor $(|0\rangle + \mathrm{e}^{2\pi \mathrm{i}\, 0.x_{n-l+1} \cdots x_n}\,|1\rangle)/\sqrt{2}$. Reindexing with $k = n - l + 1$ produces (1.4), with the factor of index k on wire $n + 1 - k$. In particular the top wire $l = 1$, which carries the most significant output bit, ends in the state $(|0\rangle + \mathrm{e}^{2\pi \mathrm{i}\, 0.x_n}\,|1\rangle)/\sqrt{2}$, whose single phase digit is the least significant input bit, and the bottom wire $l = n$ ends in $(|0\rangle + \mathrm{e}^{2\pi \mathrm{i}\, 0.x_1 x_2 \cdots x_n}\,|1\rangle)/\sqrt{2}$, whose phase carries every input bit. This is the bit-by-bit content of the wire-order convention stated after (1.4).

1.1.4 Circuit structure

The product form says that the output is a product state whose k-th factor depends only on the bits $x_k, \ldots, x_n$, through a single relative phase. The factor is produced by a Hadamard gate on qubit k, which contributes the leading digit $x_k/2$, followed by phase rotations $R_m = \mathrm{diag}(1, \mathrm{e}^{2\pi \mathrm{i}/2^m})$ on qubit k, each controlled by one of the later qubits $x_{k+1}, \ldots, x_n$, which contribute the remaining digits. Qubit k receives $n - k$ controlled rotations, so the whole circuit uses $n(n+1)/2$ gates of these two kinds, followed by $\lfloor n/2 \rfloor$ SWAP gates that undo the bit reversal. The circuit therefore uses $O(n^2)$ gates, meaning at most a constant multiple of n^2, which is $O(\log^2 N)$. The three-qubit circuit is drawn in section 1.1.5.

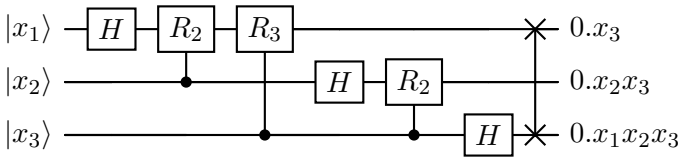
Wrong Concept 1.1: The QFT speeds up classical signal processing

The classical fast Fourier transform of a vector of N numbers costs $O(N \log N)$ operations and returns all N transformed coefficients. The quantum Fourier transform costs $O(\log^2 N)$ gates, and it is tempting to conclude that a quantum computer performs Fourier analysis exponentially faster. It does not, for two reasons. First, the data vector must be loaded into the N amplitudes of a register, and no general procedure does this in fewer than $O(N)$ steps. Second, and more fundamentally, the transformed coefficients are amplitudes, and

?? showed that amplitudes are not readable: a measurement returns a single index y, with probability equal to the squared modulus of the y-th transformed coefficient, and the state is then gone. The transform acts on the whole vector, but one obtains one sample from it, not the vector. Every algorithm in this chapter is designed so that a single sample from a Fourier-transformed state is already informative.

1.1.5 The three-qubit circuit

Let $R_m = \mathrm{diag}(1, \mathrm{e}^{2\pi \mathrm{i}/2^m})$, so that a controlled-$R_m$ with control in state $|c\rangle$ multiplies the target's $|1\rangle$ amplitude by $\mathrm{e}^{2\pi \mathrm{i}c/2^m}$. Note that $R_1 = Z$ and R_2 is the phase gate S. The Hadamard gate on qubit k produces $(|0\rangle + \mathrm{e}^{2\pi \mathrm{i}\, 0.x_k}\,|1\rangle)/\sqrt{2}$, since $\mathrm{e}^{2\pi \mathrm{i} x_k/2} = (-1)^{x_k}$; the controlled-$R_m$ from qubit $k+m-1$ then appends the digit $x_{k+m-1}/2^m$ to the binary fraction. For $n = 3$:



The labels on the right name the phase θ in the output state $(|0\rangle + \mathrm{e}^{2\pi \mathrm{i}\theta}\,|1\rangle)/\sqrt{2}$ of each wire. Immediately before the SWAP, wire k carries $0.x_k \cdots x_n$, which is the factor of index k in (1.4); the SWAP of wires 1 and 3 places $0.x_3$ on the top wire and $0.x_1x_2x_3$ on the bottom wire, as the convention requires. Each control acts on a qubit that has not yet been touched by a Hadamard gate, which is why the gates on the lower wires are delayed. The circuit uses $3 + 3 = 6$ Hadamard and rotation gates and one SWAP; in general there are $n(n+1)/2$ and $\lfloor n/2 \rfloor$ respectively. Some authors omit the SWAP gates and instead relabel the output wires in reversed order; either choice is correct provided the convention is stated.

1.2 Phase estimation

A unitary U acting on an eigenvector $|u\rangle$ multiplies it by a complex number of modulus one, $U\,|u\rangle = \mathrm{e}^{2\pi \mathrm{i}\varphi}\,|u\rangle$ with $\varphi \in [0, 1)$. We call φ the eigenphase. Applied to $|u\rangle$ alone, U produces only a global phase, which **??** showed to be unobservable. The eigenphase becomes observable when U is applied under control, because phase kickback, introduced in **??**, moves the phase onto the control qubit: for a controlled-U in the sense of **??**,

$$\text{controlled-}U : |+\rangle \otimes |u\rangle \longmapsto \frac{|0\rangle + \mathrm{e}^{2\pi \mathrm{i}\varphi}\,|1\rangle}{\sqrt{2}} \otimes |u\rangle\,. \tag{1.6}$$

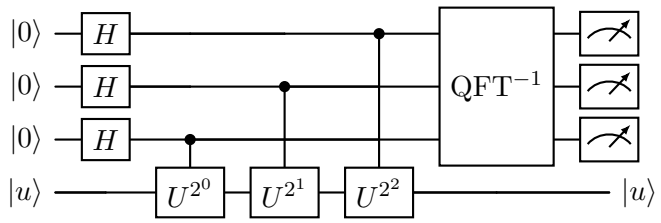
The target is unchanged, and the control now carries φ as a relative phase. Reading one relative phase to high precision from one qubit would require many repetitions. Phase estimation instead spreads the binary digits of φ over a register, in exactly the pattern that QFT^{-1} knows how to read.

Definition 1.3: Phase estimation problem

Given a unitary U, an eigenstate $|u\rangle$ with $U\,|u\rangle = \mathrm{e}^{2\pi \mathrm{i}\varphi}\,|u\rangle$, and the ability to apply the controlled operations controlled-U^{2^j} for $j = 0, 1, \ldots, t-1$, determine the eigenphase φ to t binary digits. The t control qubits form the counting register; the register holding $|u\rangle$ is the target register.

The circuit is the following. Prepare each counting qubit in $|+\rangle$ by a Hadamard gate. Apply controlled-U^{2^j} with the counting qubit of index j as control and the target register as target. Since

$U^{2^j}|u\rangle = \mathrm{e}^{2\pi\mathrm{i}\,2^j\varphi}|u\rangle$, the kickback (1.6) writes the phase $\mathrm{e}^{2\pi\mathrm{i}\,2^j\varphi}$ onto counting qubit j. Finally apply QFT^{-1} to the counting register and measure it in the computational basis. We draw the counting qubit with the largest index $j = t-1$ on the top wire, so that the top wire is the most significant bit of the result.



The figure shows $t = 3$. After the controlled operations, the counting register is in the product state whose factor on qubit j is $(|0\rangle + \mathrm{e}^{2\pi\mathrm{i}\,2^j\varphi}|1\rangle)/\sqrt{2}$, and expanding the product gives

$$\frac{1}{\sqrt{2^t}} \sum_{y=0}^{2^t-1} \mathrm{e}^{2\pi\mathrm{i}\varphi y}|y\rangle\,, \tag{1.7}$$

where $y = y_1 \cdots y_t$ is read with y_1 on the top wire. The full calculation is carried out in section 1.2.1. Comparing (1.7) with (1.1) gives the decisive statement of this lecture.

1.2.1 Exact eigenphases

Theorem 1.1 Phase estimation, exact case

Suppose the eigenphase has a binary expansion of exactly t digits, $\varphi = 0.\varphi_1\varphi_2\cdots\varphi_t$, and let $\tilde{\varphi} = \varphi_1\varphi_2\cdots\varphi_t$ be the integer with those digits, so that $\varphi = \tilde{\varphi}/2^t$. Then the counting register state (1.7) equals $\mathrm{QFT}\,|\tilde{\varphi}\rangle$, the Fourier basis state $|f_{\tilde{\varphi}}\rangle$ on t qubits. Consequently QFT^{-1} maps it to $|\varphi_1\varphi_2\cdots\varphi_t\rangle$, and the measurement returns the t digits of φ with certainty, with φ_1 on the top wire.

The proof is one line: with $\varphi = \tilde{\varphi}/2^t$ and $N = 2^t$ the phase $\mathrm{e}^{2\pi\mathrm{i}\varphi y}$ is $\mathrm{e}^{2\pi\mathrm{i}\tilde{\varphi}y/N}$, which is the coefficient in (1.1) with $x = \tilde{\varphi}$. The eigenphase was never written into any amplitude's modulus. It was written into the relative phases of the counting register, in exactly the pattern of a Fourier basis state, and the inverse transform converts that pattern into an integer.

The full calculation, with the wire order made explicit, is the following. Label the counting qubits by $j = t-1, \ldots, 1, 0$ from top to bottom, and let qubit j control U^{2^j}. After the Hadamard gates the state is $|+\rangle^{\otimes t} \otimes |u\rangle$. Each controlled-$U^{2^j}$ acts on its own control qubit and the common target, and the target is an eigenstate of every power of U, so the operations commute and each contributes the kickback (1.6) with φ replaced by $2^j\varphi$. The counting register becomes

$$\bigotimes_j \frac{|0\rangle + \mathrm{e}^{2\pi\mathrm{i}\,2^j\varphi}|1\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2^t}} \sum_{y_1,\ldots,y_t\in\{0,1\}} \mathrm{e}^{2\pi\mathrm{i}\varphi\sum_j y_{t-j}2^j}|y_1\cdots y_t\rangle = \frac{1}{\sqrt{2^t}} \sum_{y=0}^{2^t-1} \mathrm{e}^{2\pi\mathrm{i}\varphi y}|y\rangle\,, \tag{1.8}$$

where the product runs over $j = t-1, \ldots, 0$ from top to bottom, so the top wire holds the most significant bit y_1 and $y = \sum_j y_{t-j}2^j$. This is (1.7). If $\varphi = \tilde{\varphi}/2^t$ for an integer $\tilde{\varphi}$, this is $\frac{1}{\sqrt{2^t}}\sum_y \mathrm{e}^{2\pi\mathrm{i}\tilde{\varphi}y/2^t}|y\rangle = \mathrm{QFT}\,|\tilde{\varphi}\rangle$ by (1.1) with $N = 2^t$, and QFT^{-1} returns $|\tilde{\varphi}\rangle = |\varphi_1\cdots\varphi_t\rangle$. The target register is left in $|u\rangle$ throughout and can be reused.

Example 1.1 (Reading a three-bit eigenphase)

Let $t = 3$ and $\varphi = 0.101$ in binary, that is $\varphi = 5/8$. The kickback phases are $\mathrm{e}^{2\pi\mathrm{i}\cdot 4\cdot 5/8} = -1$ on the top counting qubit, $\mathrm{e}^{2\pi\mathrm{i}\cdot 2\cdot 5/8} = \mathrm{i}$ on the middle one, and $\mathrm{e}^{2\pi\mathrm{i}\cdot 5/8}$ on the bottom one. The counting register is therefore in the state $\frac{1}{\sqrt{8}}\sum_y \mathrm{e}^{2\pi\mathrm{i}\,5y/8}\,|y\rangle = \mathrm{QFT}\,|101\rangle$. Applying QFT^{-1} and measuring returns $|101\rangle$: the top wire reads $\varphi_1 = 1$, the middle wire $\varphi_2 = 0$, and the bottom wire $\varphi_3 = 1$. Under our convention the outcome, read from top to bottom, is the binary expansion of φ.

For Example 1.1, $t = 3$ and $\tilde{\varphi} = 5$: the three factors are $(|0\rangle - |1\rangle)/\sqrt{2}$, $(|0\rangle + \mathrm{i}\,|1\rangle)/\sqrt{2}$, and $(|0\rangle + \mathrm{e}^{5\pi\mathrm{i}/4}\,|1\rangle)/\sqrt{2}$, which are 0.1, 0.01, and 0.101 in the notation of the product form, in agreement with the convention that the top wire carries $0.\varphi_3$ and the bottom wire $0.\varphi_1\varphi_2\varphi_3$.

The point of the example is to identify the ideal outcome under the stated convention. Three roles should be kept distinct. The eigenphase φ is the unknown quantity, a property of U and $|u\rangle$ jointly. The counting register size t fixes the precision: the outcome is an integer between 0 and $2^t - 1$, so φ is resolved to within 2^{-t} at best. The controlled powers U^{2^j} are the price: the largest is $U^{2^{t-1}}$, so each additional digit of precision doubles the number of applications of U, unless U^{2^j} can be implemented directly at lower cost. The next section presents a unitary for which it can.

1.2.2 Inexact eigenphases

Remark 1.1 The inexact case

When φ has no t-digit expansion, the state (1.7) is not a single Fourier basis state but a superposition of them concentrated near $\tilde{\varphi} \approx 2^t\varphi$, and the measurement outcome is random. The concentration is strong: the outcome is the nearest t-digit approximation to φ with probability at least $4/\pi^2 \approx 0.405$, and to obtain an estimate within 2^{-n} of φ with probability at least $1 - \epsilon$ it suffices to take $t = n + \lceil \log_2(2 + 1/(2\epsilon)) \rceil$ counting qubits. Additional counting qubits therefore buy confidence, not only precision. The argument is outlined below, with a pointer to its complete proof.

Write $\varphi = \tilde{\varphi}/2^t + \delta$ with $\tilde{\varphi}$ the integer nearest to $2^t\varphi$ and $|\delta| \leqslant 2^{-t-1}$. Applying QFT^{-1} to (1.7) gives the amplitude of the outcome $\tilde{\varphi} + m$ as $\frac{1}{2^t}\sum_y \mathrm{e}^{2\pi\mathrm{i}y(\delta - m/2^t)}$, a geometric sum whose modulus is $\frac{1}{2^t}|\sin(\pi 2^t\Delta)/\sin(\pi\Delta)|$ with $\Delta = \delta - m/2^t$. For $m = 0$ the bound $\sin(\pi x) \leqslant \pi x$ in the denominator and $\sin(\pi x) \geqslant 2x$ for $0 \leqslant x \leqslant 1/2$ in the numerator give probability at least $4/\pi^2$ for the nearest value $\tilde{\varphi}$. Summing the tail $|m| > 2^{t-n}$ shows that the outcome is within 2^{-n} of φ with probability at least $1 - \epsilon$ when $t = n + \lceil \log_2(2 + 1/(2\epsilon)) \rceil$. The complete argument is in Nielsen and Chuang [1, Section 5.2.1] and in Kitaev's original treatment of phase estimation [2]. The homework evaluates the geometric sum in one small instance and does not require the general bound.

1.3 Order finding and factoring

Phase estimation reads the eigenphase of a unitary. Shor's insight [3] was that a problem with no apparent connection to phases, finding the period of modular exponentiation, is an eigenphase problem for a unitary whose controlled powers are cheap. We fix an integer N to be factored, with $n = \lceil \log_2 N \rceil$ bits, and an integer a with $1 < a < N$ and $\gcd(a, N) = 1$. Modular arithmetic, the greatest common divisor, and the order of an element are used here only as definitions, which are collected in the course's reference sheet; nothing beyond their definitions is assumed.

Definition 1.4: Order

The order of a modulo N is the least positive integer r such that $a^r \equiv 1 \pmod{N}$. It exists whenever $\gcd(a, N) = 1$, and the sequence $a^0, a^1, a^2, \ldots$ reduced modulo N is periodic with period r.

No classical algorithm is known that finds r in time polynomial in n; the classical reduction outlined in section 1.3.2 shows that order finding is at least as hard as factoring. Consider the modular multiplication map

$$U_a |y\rangle = |ay \bmod N\rangle , \qquad 0 \leqslant y < N, \tag{1.9}$$

on an n-qubit register, extended as the identity on the basis states $|y\rangle$ with $y \geqslant N$. Because $\gcd(a, N) = 1$, multiplication by a permutes the residues modulo N, so U_a permutes the computational basis and is unitary. Its eigenstates are obtained by placing phases along the orbit of 1 under repeated multiplication by a.

Proposition 1.1 Eigenphases of modular multiplication

Let r be the order of a modulo N. For each $s \in \{0, 1, \ldots, r-1\}$ the state

$$|u_s\rangle = \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} \mathrm{e}^{-2\pi \mathrm{i} s k / r} \left| a^k \bmod N \right\rangle \tag{1.10}$$

satisfies $U_a |u_s\rangle = \mathrm{e}^{2\pi \mathrm{i} s/r} |u_s\rangle$. The eigenphases of U_a on these states are the fractions s/r, and they carry the order r in their denominators. Moreover $\frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} |u_s\rangle = |1\rangle$.

Both statements are verified in section 1.3.1. The first turns order finding into phase estimation: estimating the eigenphase of U_a on $|u_s\rangle$ yields an approximation to s/r. The second removes what would otherwise be a fatal obstacle. Preparing $|u_s\rangle$ requires knowing r, which is what we are trying to find. But the computational basis state $|1\rangle$, which costs nothing to prepare, is a uniform superposition of all the $|u_s\rangle$. By linearity, phase estimation applied to $|1\rangle$ behaves as if a value of s were drawn uniformly at random and the corresponding $|u_s\rangle$ were supplied: the measurement returns an approximation to s/r for an unknown random s. This is the superposition trick that makes eigenstate preparation unnecessary.

The controlled powers are cheap for the reason that makes this problem special. Since $U_a^{2^j} = U_{a^{2^j} \bmod N}$, the power is itself a modular multiplication by a number that a classical computer obtains by repeated squaring. The whole sequence of controlled powers therefore implements modular exponentiation, $|x\rangle |y\rangle \mapsto |x\rangle |a^x y \bmod N\rangle$ for the counting register $|x\rangle$, and reversible arithmetic circuits perform it with a number of gates polynomial in n, roughly cubic with elementary multiplication. We name these circuits and do not construct them.

1.3.1 Eigenstates of modular multiplication

Apply U_a to (1.10). Since $U_a \left|a^k \bmod N\right\rangle = \left|a^{k+1} \bmod N\right\rangle$,

$$U_a \left|u_s\right\rangle = \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} \mathrm{e}^{-2\pi \mathrm{i} s k/r} \left|a^{k+1} \bmod N\right\rangle = \frac{1}{\sqrt{r}} \sum_{k'=1}^{r} \mathrm{e}^{-2\pi \mathrm{i} s (k'-1)/r} \left|a^{k'} \bmod N\right\rangle = \mathrm{e}^{2\pi \mathrm{i} s/r} \left|u_s\right\rangle , \tag{1.11}$$

where the term $k' = r$ has been identified with $k' = 0$ using $a^r \equiv 1$ and $\mathrm{e}^{-2\pi \mathrm{i} s r/r} = 1$. The states $|u_s\rangle$ are orthonormal for $s = 0, \ldots, r-1$, by the same geometric sum as for the Fourier basis, since the r residues $a^k \bmod N$ are distinct. Summing over s,

$$\frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} \left|u_s\right\rangle = \frac{1}{r} \sum_{k=0}^{r-1} \left(\sum_{s=0}^{r-1} \mathrm{e}^{-2\pi \mathrm{i} s k/r} \right) \left|a^k \bmod N\right\rangle = \frac{1}{r} \cdot r \left|a^0 \bmod N\right\rangle = \left|1\right\rangle , \tag{1.12}$$

because the inner sum is r for $k = 0$ and 0 otherwise. In fact the $|u_s\rangle$ are exactly the inverse Fourier transform, on r points, of the orbit states $\left|a^k \bmod N\right\rangle$, which is why the order appears in their eigenphases. Running phase estimation on the input $|1\rangle$ therefore produces, by linearity, the state $\frac{1}{\sqrt{r}} \sum_s |\text{estimate of } s/r\rangle \otimes |u_s\rangle$, and measuring the counting register yields an estimate of s/r for a uniformly random s. The eigenphase $s = 0$ gives the useless estimate 0; it occurs with probability $1/r$ and is discarded.

1.3.2 The classical reduction

The two classical steps of Shor's algorithm are recorded here for completeness; the course does not develop them, and the homework requires only the gcd arithmetic of a small instance, not the continued-fractions algorithm.

From the phase to the order. With $2n + 1$ counting qubits the measured value determines s/r as the unique fraction with denominator below N within the achieved precision, and the continued-fractions algorithm, which we name but do not develop, extracts r.

Precision and uniqueness. With $t = 2n + 1$ counting qubits, together with the few extra qubits of Remark 1.1, the measurement returns an integer $\tilde{\varphi}$ such that $\tilde{\varphi}/2^t$ lies within $1/2^{2n+1}$ of s/r with high probability. Since $r < N \leqslant 2^n$, two distinct fractions with denominators below N differ by at least $1/(rr') > 1/2^{2n}$, so s/r is the unique such fraction within the achieved precision of the measured value. The continued-fractions algorithm, a classical procedure that runs in time polynomial in n, recovers the fraction from the measured value. If s and r share a factor the algorithm returns the reduced fraction, and hence a divisor of r rather than r itself; the candidate is checked by computing $a^r \bmod N$. The probability that a random s is coprime to r is $\phi(r)/r$, which is at least $c/\log\log r$ for a constant c, so simple repetition needs $O(\log\log N)$ expected attempts; combining two candidate orders by their least common multiple succeeds with probability at least $1/4$ per pair and reduces the expected number of repetitions to a constant.

From the order to a factor. If r is even and $a^{r/2} \not\equiv -1 \pmod N$, then $\gcd(a^{r/2} \pm 1, N)$ is a nontrivial factor of N; for odd N that is not a prime power, a random a coprime to N satisfies both conditions with probability at least one half, and an a that is not coprime to N yields a factor directly as $\gcd(a, N)$.

If r is even and $a^{r/2} \not\equiv -1 \pmod N$, then N divides $a^r - 1 = (a^{r/2} - 1)(a^{r/2} + 1)$. It divides neither factor: $a^{r/2} \not\equiv 1$ because r is the least positive exponent with $a^r \equiv 1$, and $a^{r/2} \not\equiv -1$ by hypothesis. Hence $\gcd(a^{r/2} - 1, N)$ and $\gcd(a^{r/2} + 1, N)$ are both nontrivial factors of N, and each

is computed by the Euclidean algorithm in time polynomial in n. For an odd N that is not a prime power, an a chosen uniformly at random among the residues coprime to N satisfies both conditions with probability at least $1/2$; this is a fact from elementary number theory, proved in Nielsen and Chuang [1, Appendix 4], which we cite and do not prove. The excluded cases are handled classically: an even N has the factor 2, and a prime power is detected in polynomial time by extracting integer roots.

1.3.3 The instance $N = 15$

The powers of 7 modulo 15 are $7^1 = 7$, $7^2 = 49 = 4$, $7^3 = 28 = 13$, and $7^4 = 91 = 1$, so the order is $r = 4$. The eigenphases s/r are $0, 1/4, 1/2, 3/4$, each of which has an exact two-digit binary expansion, so two counting qubits would suffice for exact readout in this instance; the general choice $t = 2n + 1 = 9$ for $n = 4$ is designed for the worst case, in which r does not divide a power of two. The order $r = 4$ is even, and $a^{r/2} = 7^2 \equiv 4 \not\equiv -1 = 14 \pmod{15}$, so both conditions of the reduction hold. Then $\gcd(7^2 - 1, 15) = \gcd(3, 15) = 3$ and $\gcd(7^2 + 1, 15) = \gcd(5, 15) = 5$, and indeed $15 = 3 \times 5$. The identity $a^r - 1 = (a^{r/2} - 1)(a^{r/2} + 1)$ reads $2400 = 48 \times 50$, and 15 divides 2400 without dividing 48 or 50. For comparison, $a = 2$ has order 4 as well, with $2^2 = 4$ giving the same factors, while $a = 14 \equiv -1$ has order 2 with $14^1 \equiv -1$, which is the excluded case and yields nothing.

1.3.4 Shor's algorithm

Assembling the pieces gives Shor's algorithm: choose a at random, run phase estimation for U_a on the input $|1\rangle$, recover r by continued fractions, and compute the two greatest common divisors. Every step is polynomial in n, and the expected number of repetitions is small: constant if two candidate orders are combined by taking their least common multiple, and at most of order $\log \log N$ if one simply repeats until a candidate passes the check $a^r \equiv 1 \pmod{N}$. The two classical steps are outlined in section 1.3.2 and illustrated for $N = 15$ in section 1.3.3. Factoring, for which no polynomial-time classical algorithm is known, is solved in polynomial time on a quantum computer. The widely used RSA public-key system rests on the presumed hardness of factoring, and related systems rest on discrete logarithms, which the same mechanism attacks. **??** examines what this means for cryptography in practice, and which systems are and are not affected.

> **Remark 1.2** Hidden costs
>
> The circuit description of Shor's algorithm fits on half a page, and three costs are easy to overlook. First, the modular exponentiation circuit dominates: the QFT^{-1} is $O(n^2)$ gates, but the controlled arithmetic is larger by a further factor of n and requires several times n additional qubits. Second, controlled access to $U_a^{2^j}$ is a resource that the problem happens to supply; for a general unitary the powers cost 2^j applications each, so phase estimation offers no exponential saving over the 2^t applications of U that the precision 2^{-t} requires; its advantage over repeated single-qubit sampling is only the quadratic one from $1/\sqrt{M}$ to $1/M$ precision for M uses. Third, the algorithm assumes gates that remain error-free over a very long computation, and **??** showed that physical gates do not; the overhead of error correction multiplies every count above. **??** supplies a dated estimate of the resulting resource requirements. At the time of writing, no quantum computer capable of factoring numbers of cryptographic size exists.

Wrong Concept 1.2: Shor's algorithm has already broken RSA

The algorithm is a mathematical result about an idealized machine, and the demonstrations performed to date have factored very small numbers, in several cases with circuits simplified using prior knowledge of the answer. The resources required to factor a number of the size used in deployed RSA keys exceed the capabilities of current hardware by orders of magnitude in qubit count, gate count, and error rate. What the algorithm has broken is a presumption: the presumption that factoring is intrinsically hard, on which the long-term security of certain cryptographic systems was based. Whether and when a machine will exist that realizes the threat is a question about engineering that these notes do not attempt to answer.

?? takes this result as its starting point. It distinguishes the public-key systems whose security rests on factoring or discrete logarithms, which Shor's algorithm would break at sufficient scale, from symmetric ciphers and hash functions, for which the search algorithm of **??** offers only a quadratic speedup; and it explains why the threat motivates a migration of classical cryptography today, before any such machine exists.

1.4 Summary

The quantum Fourier transform is a unitary change of basis between the computational basis and the Fourier basis, whose states differ only in their pattern of relative phases. Its circuit has $O(n^2)$ gates because its output factorizes into a product of single-qubit states. It does not compute the classical Fourier transform of a data vector, since amplitudes cannot be read; a measurement after QFT^{-1} returns one sample. Phase estimation uses phase kickback from controlled powers U^{2^j} to write the eigenphase φ of a unitary into the relative phases of a counting register, producing exactly the Fourier basis state $\mathrm{QFT}\,|\tilde{\varphi}\rangle$ when φ has t digits, so that QFT^{-1} and measurement return those digits with certainty, and a good approximation with high probability otherwise. Order finding is an eigenphase problem for modular multiplication, whose eigenphases are s/r, whose controlled powers are modular exponentiation, and whose eigenstates need not be prepared because $|1\rangle$ is their uniform superposition. A short classical argument converts the order into a factor. The result is a polynomial-time factoring algorithm whose practical realization requires resources that do not currently exist.

Homework 4, part A, can be attempted after this lecture.

References

[1] Michael A. Nielsen and Isaac L. Chuang. *Quantum Computation and Quantum Information*. Cambridge University Press, Cambridge, 10th anniversary edition, 2010.

[2] Alexei Yu. Kitaev. Quantum measurements and the Abelian stabilizer problem, 1995. arXiv:quant-ph/9511026.

[3] Peter W. Shor. Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer. *SIAM Journal on Computing*, 26:1484–1509, 1997.